Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/12/2024
PI Name:	Boppart, Marni		
Project Title:	Design of an Extracellular Vesicle Approach to Protect Human Health in Space		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mboppart@illinois.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	
Organization Name:	University of Illinois		
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PI Address 2:	405 N. Mathews Ave, MC-251		
PI Web Page:			
City:	Urbana	**State:**	IL
Zip Code:	61801	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 TRISH Biomedical Research Advances for Space Health (BRASH)
Start Date:	07/01/2023	**End Date:**	06/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	7	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Burd, Nicholas (University Of Illinois, Urbana-Champaign) Kong, Hyunjoon (University Of Illinois, Urbana-Champaign) Rhodes, Justin (Self) Saif, M Taher (University Of Illinois, Urbana-Champaign) Sweedler, Jonathan (University Of Illinois, Urbana-Champaign)		
Grant/Contract No.:	NNX16AO69A-T0701		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Exercise has been embraced by astronauts to reduce tissue degeneration; yet diminished mechanical cues prevent full preservation of tissue mass and function during long-term spaceflight. Therefore, autologous supplementation with the exercise secretome may represent an ideal disruptive approach to improve endogenous repair. Based on our preliminary results, demonstrating the capacity for ExerVs to promote consistent and visible increases in vascular growth in murine skeletal muscle, we propose that ExerVs may be collected under normal gravitation force, cryopreserved, and reintroduced prior to and during spaceflight. We will accomplish our goal by pursuing the following two Aims: (1) to demonstrate the capacity for human-derived ExerVs to improve endogenous repair and growth mechanisms in vitro and in vivo, and (2) to demonstrate the capacity for artificially manufactured ExerVs to improve endogenous repair and growth mechanisms in vitro and in vivo. As directed by the solicitation, we will evaluate the impact of natural and artificial human-derived ExerVs on multiple organ systems, including vessels, muscle and brain, in collaboration with colleagues in Neuroscience, Mechanical Science and Engineering, and Chemical and Biomolecular Engineering. Using our innovative and integrative approach, we are confident we will deliver a novel product that will effectively preserve human health in space in a controlled and safe manner.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	At the end of the first year of funding, we have successfully collected blood samples from 20 human subjects (n=10 male and 10 female) after a 6-week supervised, progressive, multi-component training program. Increases in strength and lean body mass were observed at the end of the study. ExerVs have been isolated from plasma samples for characterization, as well as in vitro and in vivo experiments. These important experiments are now in progress.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Awards	Boppart M. "Allen Distinguished Investigator Award. " Oct-2023
Awards	Boppart M. "Saul J. Morse and Anne B. Morgan Professorship in Applied Health Sciences." May-2024
Awards	Saif T. "Member, National Academy of Engineering." Jan-2024